적분 점화식 정리

1. $\int \sin^n x dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x dx$

증명)

$$\int \sin^n x\, dx = \int \sin x \sin^{n-1}x\, dx \quad = -\sin^{n-1}x\cos x - \int (-\cos x)(n-1)\sin^{n-2}x\cos x\, dx \quad (\because \text{부분적분})$$

$$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\cos^2 x\, dx$$

$$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x(1-\sin^2 x)dx \quad (\because \sin^2 x + \cos^2 x = 1)$$

$$= -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x - (n-1)\int \sin^n x dx$$

여기서 $I = \int \sin^n x\, dx$ 라 하면 $I = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x dx - (n-1)I$ 이다.

$\Rightarrow (n-1)I + I = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x dx$

$\Rightarrow nI = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x dx$

$\Rightarrow I = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x dx$

즉, $\int \sin^n x dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x dx$

2. $\int \cos^n x\, dx = \frac{1}{n}\sin x\cos^{n-1}x + \frac{n-1}{n}\int \cos^{n-2}x\, dx$

증명)

$$\int \cos^n x\, dx = \int \cos x\cos^{n-1}x\, dx \quad = \sin x\cos^{n-1}x + (n-1)\int \sin^2 x\cos^{n-2}x dx \quad (\because \text{부분적분})$$

$$= \sin x\cos^{n-1}x + (n-1)\int (1-\cos^2 x)\cos^{n-2}x dx \quad (\because \sin^2 x + \cos^2 x = 1)$$

$$= \sin x\cos^{n-1}x + (n-1)\int \cos^{n-2}x\, dx - (n-1)\int \cos^n x\, dx$$

$$= \sin x\cos^{n-1}x + (n-1)\int \cos^{n-2}x\, dx - (n-1)\int \cos^n x\, dx$$

여기서 $I = \int \cos^n x\, dx$ 라 하면 $I = \sin x\cos^{n-1}x + (n-1)\int \cos^{n-2}x\, dx - (n-1)I$ 이다.

$\Rightarrow (n-1)I + I = \sin x\cos^{n-1}x + (n-1)\int \cos^{n-2}x\, dx$

$\Rightarrow nI = \sin x\cos^{n-1}x + (n-1)\int \cos^{n-2}x\, dx$

$\Rightarrow I = \frac{1}{n}\sin x\cos^{n-1}x + \frac{n-1}{n}\int \cos^{n-2}x\, dx$

즉, $\int \cos^n x\, dx = \frac{1}{n}\sin x\cos^{n-1}x + \frac{n-1}{n}\int \cos^{n-2}x\, dx$

3. $\int \sec^n x\, dx = \frac{1}{n-1}\tan x\sec^{n-2}x + \frac{n-2}{n-1}\int \sec^{n-2}x\, dx$

증명)

$$\int \sec^n x\, dx = \int \sec^2 x\sec^{n-2}x\, dx = \tan x\sec^{n-2}x - \int \tan x\,(n-2)\sec^{n-3}x\sec x\tan x\, dx \quad (\because \text{부분적분})$$

$$= \tan x\sec^{n-2}x - (n-2)\int \sec^{n-2}x\tan^2 x\, dx$$

$$= \tan x\sec^{n-2}x - (n-2)\int \sec^{n-2}x(\sec^2 x - 1)\, dx \quad (\because 1 + \tan^2 x = \sec^2 x)$$

$$= \tan x\sec^{n-2}x - (n-2)\int \sec^n x\, dx + (n-2)\int \sec^{n-2}x\, dx$$

여기서 $I = \int \sec^n x dx$ 라 하면 $I = \tan x\sec^{n-2}x - (n-2)I + (n-2)\int \sec^{n-2}x\, dx$

$\Rightarrow (n-2)I + I = \tan x\sec^{n-2}x + (n-2)\int \sec^{n-2}x\, dx$

$\Rightarrow (n-1)I = \tan x\sec^{n-2}x + (n-2)\int \sec^{n-2}x\, dx$

$\Rightarrow I = \frac{1}{n-1}\tan x\sec^{n-2}x + \frac{n-2}{n-1}\int \sec^{n-2}x\, dx$

즉, $\int \sec^n x\, dx = \frac{1}{n-1}\tan x\sec^{n-2}x + \frac{n-2}{n-1}\int \sec^{n-2}x\, dx$

4. $\int \csc^n x\, dx = -\frac{1}{n-1}\cot x \csc^{n-2}x + \frac{n-2}{n-1}\int \csc^{n-2}x\, dx$

증명)

$\int \csc^n x\, dx = \int \csc^2 x \csc^{n-2}x\, dx = -\cot x \csc^{n-2}x - \int \cot x\,(n-2)\csc^{n-3}x \csc x \cot x\, dx$ (∵ 부분적분)

$= -\cot x \csc^{n-2}x - (n-2)\int \cot^2 x \csc^{n-2}x\, dx$

$= -\cot x \csc^{n-2}x - (n-2)\int (\csc^2 x - 1)\csc^{n-2}x\, dx \quad (\because 1+\cot^2 x = \csc^2 x)$

$= -\cot x \csc^{n-2}x - (n-2)\int \csc^n x\, dx + (n-2)\int \csc^{n-2}x\, dx$

여기서 $I = \int \csc^n x dx$ 라 하면 $I = -\cot x \csc^{n-2}x - (n-2)I + (n-2)\int \csc^{n-2}x\, dx$

$\Rightarrow (n-2)I + I = -\cot x \csc^{n-2}x + (n-2)\int \csc^{n-2}x\, dx$

$\Rightarrow (n-1)I = -\cot x \csc^{n-2}x + (n-2)\int \csc^{n-2}x\, dx$

$\Rightarrow I = -\frac{1}{n-1}\cot x \csc^{n-2}x + \frac{n-2}{n-1}\int \csc^{n-2}x\, dx$

즉, $\int \csc^n x\, dx = -\frac{1}{n-1}\cot x \csc^{n-2}x + \frac{n-2}{n-1}\int \csc^{n-2}x\, dx$

5. $\int \tan^n x\, dx = \frac{1}{n-1}\tan^{n-1}x - \int \tan^{n-2}x\, dx$

증명)

$\int \tan^n x\, dx = \int \tan^2 x \tan^{n-2}x\, dx = \int (\sec^2 x - 1)\tan^{n-2}x\, dx \quad (\because \tan^2 x = \sec^2 x - 1)$

$= \int \sec^2 x \tan^{n-2}x\, dx - \int \tan^{n-2}x\, dx$

$= \int t^{n-2}dt - \int \tan^{n-2}x\, dx \quad (\because \tan x = t)$

$= \frac{1}{n-1}t^{n-1} - \int \tan^{n-2}x\, dx$

$= \frac{1}{n-1}\tan^{n-1}x - \int \tan^{n-2}x\, dx$

즉, $\int \tan^n x\, dx = \frac{1}{n-1}\tan^{n-1}x - \int \tan^{n-2}x\, dx$

6. $\int \cot^n x\, dx = -\frac{1}{n-1}\cot^{n-1}x - \int \cot^{n-2}x\, dx$

증명)

$\int \cot^n x\, dx = \int \cot^2 x \cot^{n-2}x\, dx = \int (\csc^2 x - 1)\cot^{n-2}x\, dx \quad (\because 1+\cot^2 x = \csc^2 x)$

$= \int \csc^2 x \cot^{n-2}x\, dx - \int \cot^{n-2}x\, dx$

$= -\int t^{n-2}dt - \int \cot^{n-2}x\, dx \quad (\because \cot x = t)$

$= -\frac{1}{n-1}t^{n-1} - \int \cot^{n-2}x\, dx$

$= -\frac{1}{n-1}\cot^{n-1}x - \int \cot^{n-2}x\, dx$

즉, $\int \cot^n x\, dx = -\frac{1}{n-1}\cot^{n-1}x - \int \cot^{n-2}x\, dx$